Date: 28/03/2026

To
Md. Hefzul Hossain Papon
Lecturer,
Operating System Lab,
Daffodil International University.

Subject: Request for Permission to Proceed with LifeLine-RxGuard – A Hospital Integrated Pharmacy Management System.

Dear sir,

We, as a team, would like to propose a new project**, LifeLine-RxGuard – A Hospital Integrated Pharmacy Management System**, under your supervision for this semester.

Our project focuses on developing a secure and efficient hospital integrated and pharmacy management system using **Bash Shell Scripting along with Zenity GUI** dialogs in a **Linux environment**. The system will manage **medicine inventory, billing, blood bank records, user authentication, and audit tracking** using a flat-file database approach.

Through this project, we aim to strengthen our understanding of system shall scripting, file-based data management, authentication mechanisms, and real-world healthcare workflow automation.

We kindly request your permission to proceed with this project. Your approval will be highly appreciated for its successful implementation.

Thank you for your time and consideration.

Sincerely,

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Team Members:

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LifeLine-RxGuard – A Hospital Integrated Pharmacy Managements System.

Introduction

A **Hospital Integrated Pharmacy Management System** is a lightweight healthcare management solution developed **using Bash shell scripting in a Linux environment**. The system is designed to automate pharmacy-related tasks inside a hospital, including **medicine inventory management, billing operations, blood bank monitoring, and system user authentication.**

The system integrates pharmacy activities with hospital operations to ensure accurate medicine tracking and proper record management. It helps pharmacists and hospital staff monitor **medicine availability, manage stock levels, generate patient bills, and maintain transaction logs.**

This system uses structured **text files (.txt)** to store data. These files act as a flat-file database and are managed using Linux commands and shell scripts. This approach makes the system lightweight, easy to install, and suitable for small clinics or hospitals that have limited technical infrastructure.

Problem Statement

Many small hospitals and pharmacies still use manual record-keeping methods to manage medicine inventory and billing records. This often leads to several challenges, including

- Errors in manual data entry
- Difficulty tracking medicine stock and expiry dates
- Inefficient billing processes
- Lack of proper user authentication and security
- Anyone can use meds without prescription which should not be allowed

Additionally, many existing hospital management systems require complex infrastructure such as database servers and web servers, which may not be feasible for small healthcare organizations.

To address these challenges, this project proposes a simple and cost-effective pharmacy management system that operates within a Linux environment and stores all operational data using structured text files. This allows hospitals to manage pharmacy operations efficiently without relying on complex software systems.

Objectives

- The main objectives of this project are:
- Develop a hospital integrated pharmacy management system using Bash shell scripting.
- Implement a secure login and user authentication system.
- Manage medicine inventory including stock updates and expiry tracking.
- Generate patient billing and invoice records.
- Implement a blood bank management module.
- Maintain system activity logs for monitoring operations.
- Provide a simple graphical user interface using Zenity dialogs.
- Implement data backup mechanisms to prevent data loss.
- Reduce errors and improve efficiency in pharmacy record management.

Scope

Included:

- User authentication and role management
- Medicine inventory management (Add, Update, Change the expiry date, Search)
- Low-stock and expiry alerts
- Billing and invoice system
- Blood bank management
- Audit trail logging
- Backup system
- System activity logging and monitoring
- Data backup and recovery system
- Graphical interaction through Zenity dialogs
- Check expiry date

Excluded:

Use .txt fill to Read and write.

Methodology

- **System Design:** Plan workflow for pharmacy, billing, and blood bank modules.
- **Data Storage Design:** Create structured flat-file database system using .txt files.
- **Authentication Module:** Implement role-based access with hashed password storage.
- **GUI Development:** Use Zenity to create interactive forms, alerts, and dashboards.
- **Script Development:** Implement logic using Bash with tools like grip, Echo cut and sed.
- **Testing:** Perform functionality testing for inventory, billing, authentication, and error handling.
- **Documentation:** Prepare user manual and technical documentation.

Expected Outcomes

- A functional hospital integrated pharmacy management system running on Linux.
- Secure authentication with role-based access control.
- Efficient medicine stock and expiry management.
- Automated billing and invoice generation.
- Organized blood bank management system.
- Structured audit logs for monitoring activities.
- Complete documentation and demonstration.

Conclusion

The **Hospital Integrated Pharmacy Management System** is designed to simplify pharmacy operations in a hospital by using a lightweight solution based on **Bash shell scripting and Linux utilities**. The system integrates important functions such as medicine inventory management, billing generation, blood bank tracking, and user authentication to help hospital staff manage records efficiently.

This project uses **text files (.txt) as a flat-file database**, making the system simple, portable, and easy to maintain. Through this project, we will gain practical experience in **shell scripting, file handling, and Linux system tools**, which are important concepts in operating systems.

Overall, the system demonstrates how Linux-based tools can be used to automate healthcare management tasks while reducing manual errors and improving efficiency. It also provides a strong foundation for future improvements such as database integration and web-based systems